

A CASE-STUDY BASED REPORT
ON
ESPORTS MATCH OUTCOME PREDICTION USING DEEP LEARNING: A DATA-DRIVEN CASE STUDY

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ABSTRACT

Esports has emerged as a rapidly growing competitive industry, generating large volumes of data from player performance, team statistics, match history, and in-game events. Predicting the outcome of esports matches is challenging due to the dynamic nature of gameplay and the influence of multiple performance factors. This project presents a deep learning-based approach for esports match outcome prediction, using historical match data and relevant player and team statistics as input features. The proposed system applies data preprocessing, feature selection, and deep neural network techniques to identify patterns associated with winning and losing outcomes. The model is trained and evaluated on historical esports match datasets, with performance measured through accuracy, precision, recall, F1-score, and other classification metrics. By learning complex relationships within the data, the proposed approach aims to provide more reliable match predictions than traditional statistical methods. The system can assist players, analysts, coaches, and esports enthusiasts in understanding performance trends and estimating the likely outcome of future matches. Overall, this study demonstrates the potential of deep learning for data-driven decision-making and predictive analytics in the rapidly evolving esports domain.

1. INTRODUCTION

1.1 Background

Esports, or electronic sports, has developed into a major competitive and entertainment industry, attracting millions of players and viewers worldwide. Professional esports competitions involve highly skilled teams and players who compete in games such as multiplayer online battle arenas (MOBAs), first-person shooters (FPS), and real-time strategy games. The outcome of an esports match can depend on several factors, including previous match performance, player statistics, team coordination, game strategies, and individual player skills. Because of these complex and dynamic factors, accurately predicting match outcomes is a challenging task.

The rapid growth of esports has also resulted in the availability of large amounts of historical match data. This data contains information such as team win rates, player performance, rankings, match history, game statistics, and other relevant features. Machine learning techniques can analyze these datasets and identify patterns that may not be easily recognized through traditional analysis. Among these techniques, deep learning is particularly useful because it can learn complex relationships and patterns from large datasets with minimal manual intervention.

1.2 Motivation for the Project

Traditional statistical methods rely on simple relationships between a few variables and struggle to capture the non-linear, multi-factor nature of competitive gameplay. With the growing availability of historical match data, there is a clear opportunity to apply deep neural networks that can learn such relationships automatically. Reliable predictions are valuable to several groups:

- Players and coaches, for understanding strengths, weaknesses and performance trends.
- Analysts and researchers, for data-driven insight into the factors that influence winning.
- Teams and esports enthusiasts, for estimating the likely outcome of upcoming matches.

1.3 Overview of the Work Carried Out

The project "Esports Match Outcome Prediction Using Deep Learning" focuses on developing a predictive model capable of estimating the outcome of esports matches based on historical and performance-related data. The collected data is first cleaned and preprocessed to remove inconsistencies and convert relevant information into suitable numerical features. A deep learning model is then trained using these features to classify matches according to their expected outcomes, such as win or loss. The performance of the proposed model is evaluated using metrics such as accuracy, precision, recall, F1-score, and the confusion matrix.

1.4 Scope and Responsible-Use Statement

The scope of this study is limited to binary classification (win or loss) of esports matches using historical, performance-related data. The system provides data-driven insight into team performance and potential match outcomes and is intended for analytical, educational and research purposes. Predictions are probabilistic estimates, not guarantees, and must not be used as the sole basis for betting, gambling or any financial decision. Any player or team data used should be publicly available or properly authorised, and individual players should not be profiled or judged solely on model output.

2. PROBLEM STATEMENT

2.1 Case-Study Scenario

An esports analytics team has access to a large archive of historical match records containing team win rates, player performance figures, rankings, match history and in-game statistics. Coaches and analysts want a dependable way to estimate the likely result of an upcoming match from this information, instead of relying only on intuition or simple averages.

2.2 Central Case-Study Question

Can a deep learning model trained on historical player and team statistics predict the outcome (win or loss) of an esports match more reliably than traditional statistical methods?

2.3 Formal Problem Definition

Given a feature vector x describing a match (team and player statistics, rankings and match history) and a binary label y in $\{0, 1\}$ indicating loss or win, the task is to learn a function $f(x)$ that maps the input features to the predicted outcome so that the prediction error on unseen matches is minimised. The learned model is assessed using accuracy, precision, recall, F1-score and the confusion matrix.

2.4 Objectives that Guide the Study

- To collect and understand historical esports match data.
- To clean and preprocess the data and select the most relevant features.
- To design and train a deep neural network for win/loss classification.
- To evaluate the model using accuracy, precision, recall, F1-score and the confusion matrix.
- To compare its reliability with traditional statistical approaches.
- To present the findings as data-driven insight for players, coaches, analysts and enthusiasts.

2.5 Assumptions and Limitations

- Historical match data is assumed to be representative of future matches.
- Past performance is assumed to carry predictive information about match outcomes.
- Factors that cannot be captured numerically (team morale, last-minute roster changes, game patches) are not modelled.
- Model performance depends on the quality, size and balance of the available dataset.
- Results may not generalise across different games or game versions without retraining.

3. DATASET AND EXPLORATORY DATA ANALYSIS

3.1 About the Dataset

The study uses a historical esports match dataset in CSV/Excel format, loaded with Pandas and NumPy. Each record represents one match and contains information such as team win rates, player performance, rankings, match history and other game statistics, together with the match result (win or loss) as the target variable.

3.2 Purpose of EDA

Exploratory Data Analysis (EDA) is carried out to understand the structure and quality of the data before modelling. It helps to detect missing values, duplicates and outliers, examine the

distribution of features and of the win/loss classes, and discover relationships between features and the match outcome that guide feature selection.

3.3 Preparation Steps Actually Performed

- Loading the dataset and inspecting its shape, data types and summary statistics.
- Removing duplicate records and handling missing or inconsistent values.
- Converting relevant information (including categorical fields) into numerical features.
- Scaling or normalising numerical features for neural network training.
- Selecting the most relevant features.
- Splitting the data into training and testing sets.

3.4 EDA Exploration - Implementation

EDA is implemented in Python using Pandas and NumPy for data handling and Matplotlib and Seaborn for visualisation, in a Jupyter Notebook or Google Colab environment. Typical analyses include histograms and box plots of key features, a count plot of win versus loss classes, and a correlation heatmap showing how strongly each feature is related to the match outcome.

3.5 Summary of EDA Findings

EDA is used to confirm data quality, check whether the win and loss classes are balanced, and identify the features that show the strongest relationship with winning, such as team win rate, player performance and ranking. These observations determine which features are retained as inputs to the deep learning model.

4. ALGORITHM SELECTION AND IMPLEMENTATION

A deep neural network (a multilayer feed-forward network) is selected because it can learn complex, non-linear relationships among many performance features with minimal manual feature engineering, which traditional statistical methods cannot do well. The model is built with TensorFlow/Keras or PyTorch, with Scikit-learn used for splitting data, scaling and computing metrics.

4.1 Implementation Steps

- Data preprocessing and feature selection as described in Section 3.
- Defining the network: an input layer matching the number of features, several fully connected hidden layers with non-linear activation, and a sigmoid output layer giving win probability.
- Training with binary cross-entropy loss and an optimiser such as Adam on the training set.

- Predicting outcomes on the unseen test set.
- Evaluating with accuracy, precision, recall, F1-score and the confusion matrix.

4.2 Software Requirements

Software	Requirement
Operating System	Windows 10/11, Linux, or macOS
Programming Language	Python 3.9 or above
IDE / Development Environment	Jupyter Notebook, Google Colab, or Visual Studio Code
Deep Learning Framework	TensorFlow/Keras or PyTorch
Data Processing	Pandas, NumPy
Data Visualization	Matplotlib, Seaborn
Machine Learning	Scikit-learn
Database / Storage	CSV/Excel files or MySQL, if required
Web Browser	Google Chrome, Microsoft Edge, or Firefox
Version Control	Git/GitHub, optional
Internet Connection	Required for dataset collection, Google Colab, and online resources

5. CONCLUSION

This project demonstrates how artificial intelligence and deep learning can be applied to esports analytics. By utilising historical data and advanced predictive techniques, the proposed system aims to improve the understanding of factors influencing esports match outcomes and to provide a reliable framework for future research in sports and gaming analytics. Such a system can support players, coaches, teams, researchers and fans by providing data-driven insights into team performance and potential match outcomes.

Future work may include larger and richer datasets, real-time in-game features, sequence models such as LSTMs, and extension to additional games.